Incrimination of Aedes aegypti and Aedes albopictus as vectors of dengue virus serotypes 1, 2 and 3 from four states of Northeast India.

PMID: 33005866 | Indexed in PubMed

Dengue is an important vector borne disease with a great public health concern worldwide. Northeast India has experienced dengue almost every year for a decade. As studies on dengue vectors from this region are limited, we undertook an investigation to detect natural infection of the dengue virus (DENV) in potential dengue vectors of this region. Adult Aedes mosquitoes which were collected were subjected to RT-PCR for detection of infecting dengue serotype. Minimum infection rate was also determined for each positive pool. Out of the total 6229 adult Aedes mosquitoes collected, Aedes aegypti (63.3%) was abundant in comparison to Aedes albopictus (36.7%). These specimens (515 mosquito pools) were subjected to RT-PCR for detection of DENV-1, 2, 3 and 4. RT-PCR revealed the existence of DENV in both male as well as female mosquito pools suggesting natural transovarial transmission of DENV in this region. A total of 54 pools tested were positive for DENV-1, 2, 3 serotypes. This study revealed the occurence of DENV in both the potential dengue vectors from this region along with evidence of transovarial transmission which helps in persistence of the virus in nature.

[Aedes aegypti, vector of the dengue virus: spatio-temporal structure of its genetic variation].

PMID: 12669664 | Indexed in PubMed

Aedes aegypti is the main vector of dengue viruses. Methods for limiting the spread of dengue outbreaks are currently based on vector control. Estimates of population genetic organization and gene flow with respect to vector capacity have provided great insights into dengue epidemiology. In Vietnam, dengue hemorrhagic fever was detected in the 1950's and becomes today the major problem of public health. Among factors influencing dengue epidemiology, ecological disturbances have a direct impact on vectorial system functioning. Human activities through urbanization creating sanitary conditions are convenient to the vector proliferation and then, to dengue endemisation. To assess the role of the vector in the changing pattern of dengue in South-East Asia, we studied the genetic differentiation and the vector competence towards dengue 2 virus at two scales: at a spatial level (a local scale (i.e., Ho Chi Minh City) and a regional scale (i.e., Cambodia, Thailand and South Vietnam) and at a temporal scale. This study allows to propose a model of Ae. aegypti population functioning according space and time. Dynamics of dengue virus diffusion in relation with the vector, depend on the population genetic composition and its evolution.

Mosquito-infecting virus Espirito Santo virus inhibits replication and spread of dengue virus.

PMID: 33219544 | Indexed in PubMed

The primary vector of dengue virus (DENV) is Aedes aegypti. The mosquito-infecting virus, Espirito Santo virus (ESV), does not infect Vero (mammalian) cells and grows in mosquito (C6/36) cells without cytopathic effects. Effects of ESV infection on replication of DENV were explored in vitro and in vivo, analyzing protein, RNA genome expression, and plaque formation. ESV and DENV simultaneous coinfection did not block protein synthesis from either virus but did result in inhibition of DENV replication in mosquito cells. Furthermore, ESV superinfected with DENV resulted in inhibition of DENV replication and spread in A. aegypti, thus reducing vector competence. Tissue culture experiments on viral kinetics of ESV and DENV coinfection showed that neither virus significantly affects the replication of the other in Vero, HeLa, or HEK cells. Hence, ESV blocks DENV replication in insect cells, but not the mammalian cells evaluated here. Our study provides new insights into ESV-induced suppression of DENV, a globally important pathogen impacting public health.

Vector dynamics and transmission of dengue virus: implications for dengue surveillance and prevention strategies: vector dynamics and dengue prevention.

PMID: 19802582 | Indexed in PubMed

Accounting for variation in mosquito vector populations will improve dengue surveillance and prevention. Because Aedes aegypti, the principle dengue virus (DENV) vector, transmit the virus with remarkable efficiency, entomological thresholds are especially low. Assessing risk of human infection based on immature mosquito indices has proven difficult. Greater emphasis should be placed on relative abundance of adult vectors in relation to human serotype-specific herd immunity, introduction of unique viruses, mosquito-human contact and weather. The most appropriate spatial scale for assessing entomological risk is the individual household. The scale for measuring DENV transmission risk has yet to be determined but is clearly larger than the household and likely to exceed several city blocks. Because households are expected to be a primary site for human DENV infection, intradomicile vector control strategies should be a priority, especially when the force of transmission is high. The most effective intervention strategy will combine vector control with vaccine delivery for rapid and sustained disease prevention.

The dengue vector Aedes aegypti: what comes next.

PMID: 20096802 | Indexed in PubMed

Aedes aegypti is the urban vector of dengue viruses worldwide. While climate influences the geographical distribution of this mosquito species, other factors also determine the suitability of the physical environment. Importantly, the close association of A. aegypti with humans and the domestic environment allows this species to persist in regions that may otherwise be unsuitable based on climatic factors alone. We highlight the need to incorporate the impact of the urban environment in attempts to model the potential distribution of A. aegypti and we briefly discuss the potential for future technology to aid management and control of this widespread vector species.

Aedes albopictus--a new disease vector for Europe?

PMID: 8277461 | Indexed in PubMed

Aedes albopictus is an important and widespread vector of dengue in many tropical countries. Its eggs and larvae are readily transported in commodities, such as old vehicle tyres, to many temperate regions,

where it is able to survive and breed. It could thus become a serious health threat in Europe.

Vector competence of Aedes albopictus field populations from Reunion Island exposed to local epidemic dengue viruses.

PMID: 39298440 | Indexed in PubMed

Dengue virus (DENV) is the most prevalent mosquito-borne Flavivirus that affects humans worldwide. Aedes albopictus, which is naturally infected with the bacteria Wolbachia, is considered to be a secondary vector of DENV. However, it was responsible for a recent DENV outbreak of unprecedented magnitude in Reunion Island, a French island in the South West Indian Ocean. Moreover, the distribution of the cases during this epidemic showed a spatially heterogeneous pattern across the island, leading to questions about the differential vector competence of mosquito populations from different geographic areas. The aim of this study was to gain a better understanding of the vector competence of the Ae. albopictus populations from Reunion Island for local DENV epidemic strains, while considering their infection by Wolbachia. Experimental infections were conducted using ten populations of Ae. albopictus sampled across Reunion Island and exposed to three DENV strains: one strain of DENV serotype 1 (DENV-1) and two strains of DENV serotype 2 (DENV-2). We analyzed three vector competence parameters including infection rate, dissemination efficiency and transmission efficiency, at different days post-exposition (dpe). We also assessed whether there was a correlation between the density of Wolbachia and viral load/vector competence parameters. Our results show that the Ae. albopictus populations tested were not able to transmit the two DENV-2 strains, while transmission efficiencies up to 40.79% were observed for the DENV-1 strain, probably due to difference in viral titres. Statistical analyses showed that the parameters mosquito population, generation, dpe and area of sampling significantly affect the transmission efficiencies of DENV-1. Although the density of Wolbachia varied according to mosquito population, no significant correlation was found between Wolbachia density and either viral load or vector competence parameters for DENV-1. Our results highlight the importance of using natural mosquito populations for a better understanding of transmission patterns of dengue.

Complexity in the dengue spreading: A network analysis approach.

PMID: 37549129 | Indexed in PubMed

In an increasingly interconnected society, preventing epidemics has become a major challenge. Numerous infectious diseases spread between individuals by a vector, creating bipartite networks of infection with the characteristics of complex networks. In the case of dengue, a mosquito-borne disease, these infection networks include a vector-the Aedes aegypti mosquito-which has expanded its endemic area due to climate change. In this scenario, innovative approaches are essential to help public agents in the fight against the disease. Using an agent-based model, we investigated the network morphology of a dengue endemic region considering four different serotypes and a small population. The degree, betweenness, and closeness distributions are evaluated for the bipartite networks, considering the interactions up to the second order for each serotype. We observed scale-free features and heavy tails in the degree distribution and betweenness and quantified the decay of the degree distribution with a q-Gaussian fit function. The simulation results indicate that the spread of dengue is primarily driven by human-to-human and human-to-mosquito interaction, reinforcing the importance of controlling the vector to prevent episodes of epidemic outbreaks.

The reemergence of dengue virus in Sudan.

PMID: 37473544 | Indexed in PubMed

Dengue fever (DF) is a mosquito-transmitted arboviral disease caused by 1 of 4 closely related but antigenically distinct serotypes of dengue virus (DENV), DENV-1-4. The primary vector of DENV is Aedes aegypti and Aedes albopictus mosquitoes. Humans are the main carrier of the virus and the amplifying host with non-human primates plays a considerable role in sylvatic cycle. On November 8, 2022, an outbreak of dengue fever has killed at least five people in North Kordofan State. On 23 Nov 2022, the Sudanese Ministry of Health reported 3326 cases of dengue fever across 8 Sudanese States; while 23 patients died from the fever. Sudan is witnessing its worst outbreak of dengue fever in over a decade, especially in North and South Kordofan and Red Sea State are hit hard. In this review, we will focus on the recent outbreak of dengue fever in many Sudanese states.

Origin of the dengue fever mosquito, Aedes aegypti, in California.

PMID: 25077804 | Indexed in PubMed

Dengue fever is among the most widespread vector-borne infectious diseases. The primary vector of dengue is the Aedes aegypti mosquito. Ae. aegypti is prevalent in the tropics and sub-tropics and is closely associated with human habitats outside its native range of Africa. While long established in the southeastern United States of America where dengue is re-emerging, breeding populations have never been reported from California until the summer of 2013. Using 12 highly variable microsatellite loci and a database of reference populations, we have determined that the likely source of the California introduction is the southeastern United States, ruling out introductions from abroad, from the geographically closer Arizona or northern Mexico populations, or an accidental release from a research laboratory. The power to identify the origin of new introductions of invasive vectors of human disease relies heavily on the availability of a panel of reference populations. Our work demonstrates the importance of generating extensive reference databases of genetically fingerprinted human-disease vector populations to aid public health efforts to prevent the introduction and spread of vector-borne diseases.